Enterprise Server Deployment and Operating System Installation

Week 3 Portfolio Project
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Section: BSIT 4E
Date: 8/15/2026

Project Overview

This guide walks through deploying an Ubuntu Server virtual machine for a startup company. It covers the installation, the initial configuration, verifying that everything works, and documenting the process so another system administrator could follow it and reproduce the setup.

Objectives

- Install Ubuntu Server in a virtual machine.
- Configure hostname, networking, storage, and SSH.
- Verify that the server is functioning correctly.
- Document the deployment process professionally.

Software Requirements

Software	**Purpose**
Oracle VirtualBox or VMware	Virtualization
Ubuntu Server LTS ISO	Operating System
Windows Server Evaluation ISO	Bring-home activity

Virtual Machine Specifications

Component	**Specification**
Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 Virtual Processors
Storage	40 GB
Network	NAT
Hostname	server01

Ubuntu Server Installation Procedure

1. Create a new virtual machine named Ubuntu-Server-Week03.

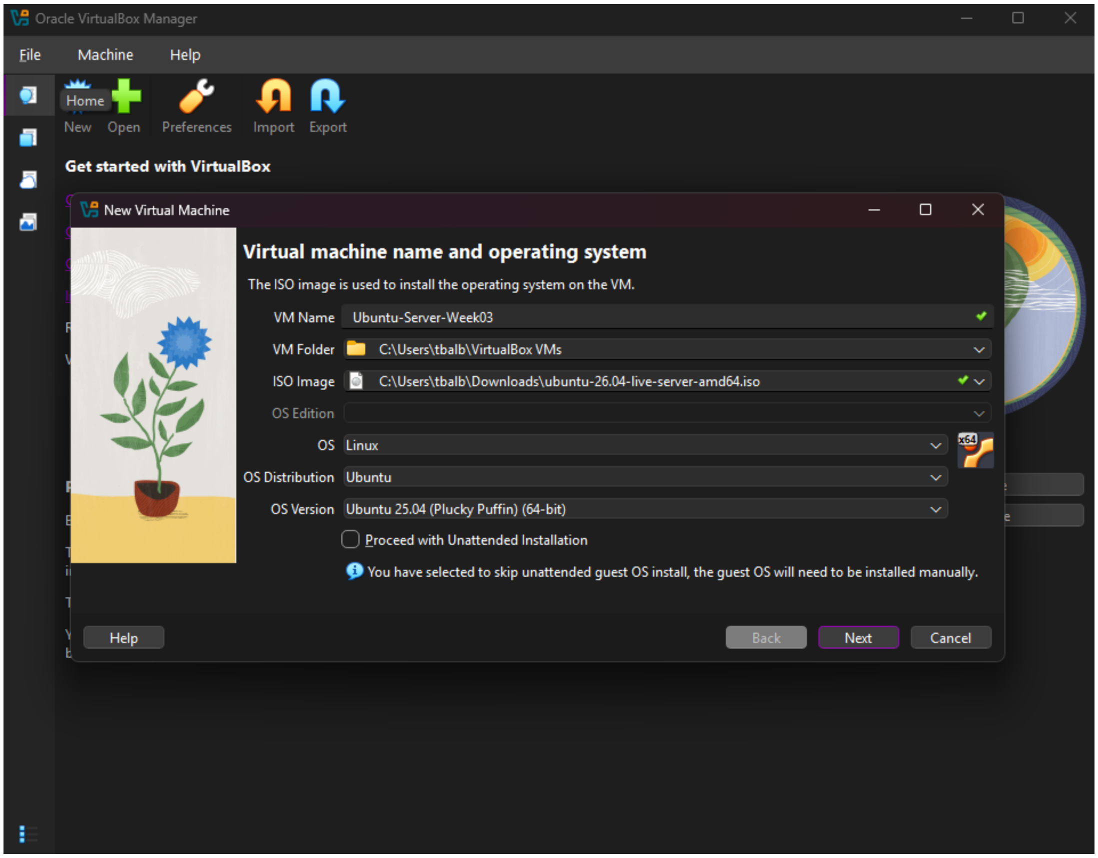

Figure 1. Ubuntu Server Creation

2. Allocate 4 GB RAM and 2 virtual processors.

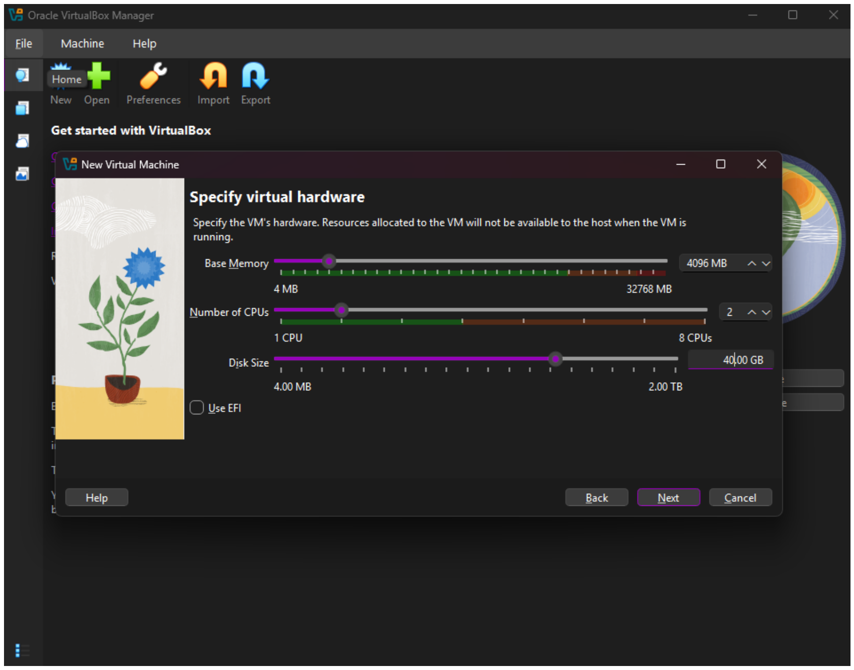

Figure 2. Ubuntu Server Hardware Configuration

3. Attach the Ubuntu Server LTS ISO and boot the VM.

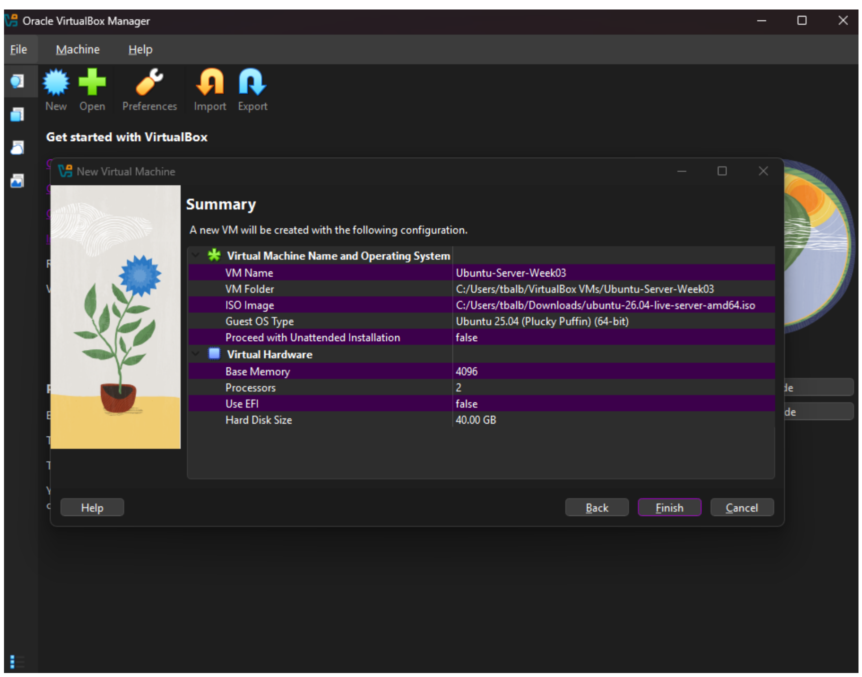

Figure 3. Ubuntu Server Summary

4. Select English language and English (US) keyboard layout.

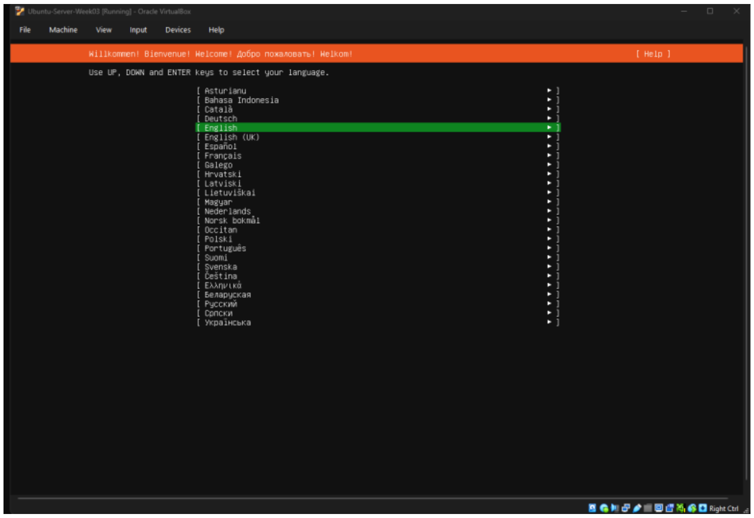

Figure 4.1. Ubuntu Server Language Selection

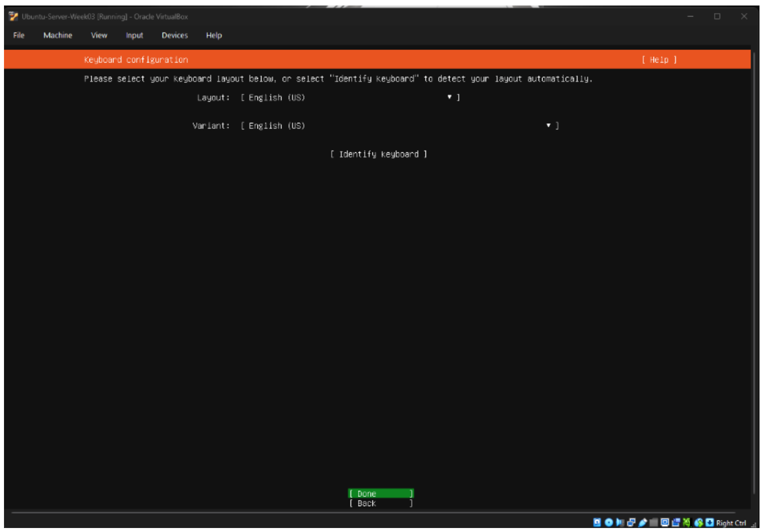

Figure 4.2. Ubuntu Server Keyboard Layout Selection

5. Accept DHCP for network configuration.

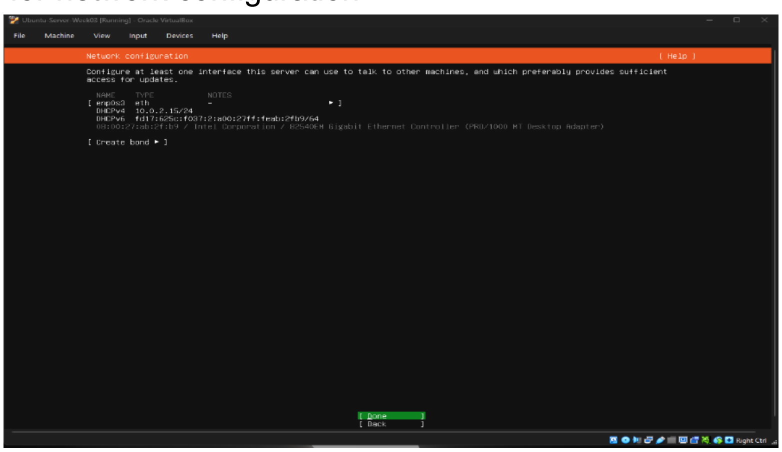

Figure 5. Ubuntu Server Network Configuration

6. Set the hostname to server01.

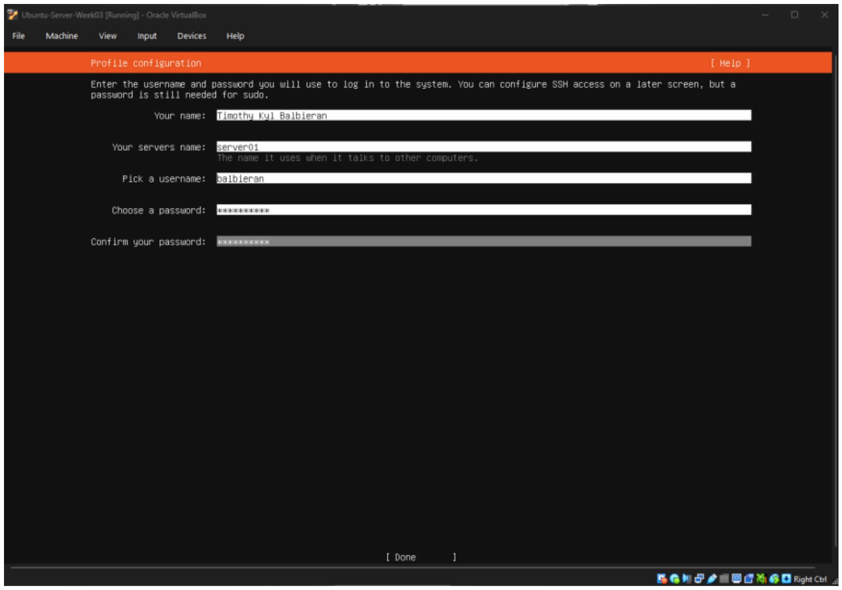

Figure 6. Ubuntu Server Profile Configuration

7. Create a non-root administrator account.
8. Use Guided, Use Entire Disk for storage.
9. Enable Install OpenSSH Server.

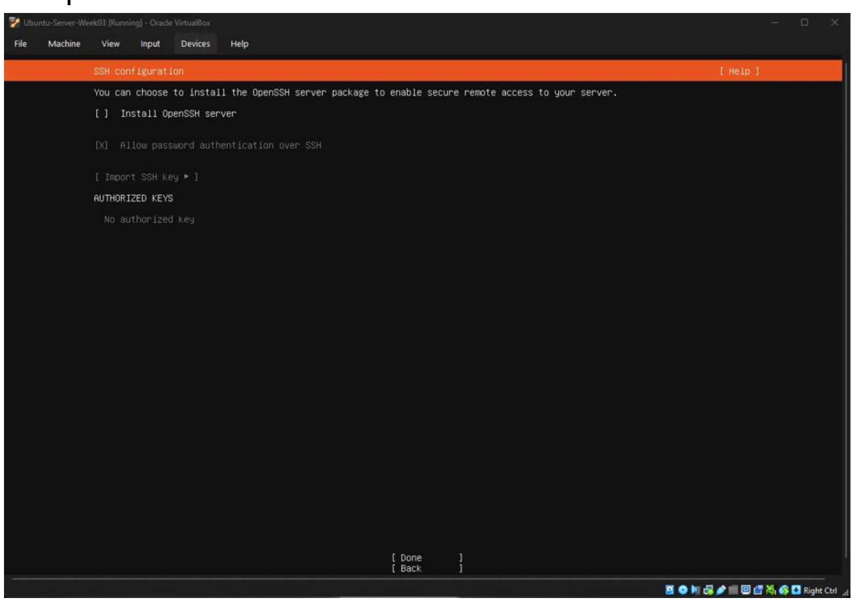

Figure 7. Ubuntu Server SSH Configuration

10. Complete the installation and restart the virtual machine.

Initial Configuration

Username: balbieran
Network Interface: enp0s3
Assigned IPv4 Address: 10.0.2.15/24

Network Configuration: DHCP
Network Mode: NAT
File System: ext4

Server Verification

Command	Purpose	Screenshot
hostname	Verify hostname	19-hostname-verification
ip addr	View IP address	20-ip-address-verification
ping -c 4 google.com	Test internet connectivity	21-ping-connectivity-test
sudo apt update && sudo apt upgrade -y	Update system	22-system-update
systemctl status ssh	Verify SSH service	23-ssh-service-status

BIOS vs UEFI Comparison

Feature	BIOS	UEFI
Definition	Basic Input/Output System. Firmware built into the motherboard that starts up the hardware and passes control to the operating system.	Unified Extensible Firmware Interface. A newer type of firmware that replaces BIOS and works more like a small operating system itself.
Boot Process	Boots use the Master Boot Record (MBR), which is a small section at the start of the disk that points to the bootloader.	Boots using the GUID Partition Table (GPT) and can load the OS directly from the EFI System Partition.
Disk Support	Can only handle disks up to 2 TB because of how MBR addresses sectors.	Can support disks way bigger than 2 TB since GPT uses 64-bit addressing.
Partition Style	MBR, which only allows up to 4 primary partitions.	GPT, which allows up to 128 partitions.
Security	No real security built in, so it's easier for malware to attack the boot process.	Has Secure Boot, which checks that the bootloader and OS haven't been tampered with before letting the system boot.
Speed	Boots slower since it loads hardware one piece at a time in an old 16-bit mode.	Boots faster because it can initialize hardware at the same time and runs in 32-bit or 64-bit mode.
Advantages	Simple and works on old hardware.	Faster boots, supports big drives, more secure, and has a nicer setup menu you can navigate with a mouse.

Disadvantages	Slow, can't handle large drives, no security features, and the interface is just plain text.	A bit more complex, and old operating systems might not even support it.
Modern Usage	Barely used anymore, mostly just on old machines or legacy systems.	Basically, the standard now on all new computers and servers, including the VM used for this project.

Ubuntu Boot Process Flowchart

Power On → BIOS/UEFI → Boot Device Detection → GRUB Boot Loader → Linux Kernel → init/systemd → Services Start → Login Prompt

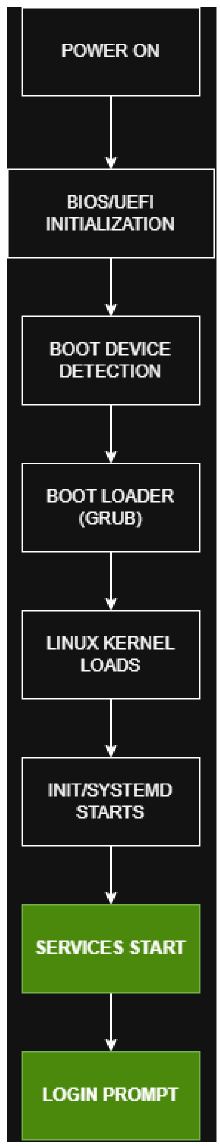

Figure 8. Ubuntu Boot Process Flowchart

Windows Server Installation Summary

For the bring-home activity, Windows Server 2025 Standard Evaluation (Desktop Experience) was installed in a separate VirtualBox virtual machine to compare against the Ubuntu Server deployment. The VM was configured with 4 GB RAM, 2 virtual processors, and a 40 GB virtual disk, matching the Ubuntu VM specs for a fair comparison. Setup used the graphical installer, selecting a clean install onto the unallocated 40 GB disk with default partitioning. During the out-of-box setup, a password was set for the built-in Administrator account, and the computer name was changed from the auto-generated WIN-4QO74L2FN8G to WinServer01 through Server Manager's Local Server properties, which required a restart to apply. After the restart, login was verified using the Administrator account, and Server Manager confirmed the new computer name and OS version (Windows Server 2025 Standard Evaluation). No major issues were encountered during this installation.

Operating System Comparison

Category	Windows	Ubuntu	Rocky
Licensing	Paid requires a product license (or evaluation copy for testing). Licensing is per-core or per-CAL depending on edition.	Free and open source. No licensing cost, though paid support (Ubuntu Pro) is optional.	Free and open source, built as a downstream rebuild of Red Hat Enterprise Linux (RHEL) with no license fee.
Interface	Has a full graphical desktop by default (though a Server Core, command-line-only option exists too). Very approachable for people coming from regular Windows.	Command-line only by default (no GUI on Server edition). Managed through the terminal or remotely through SSH.	Same as Ubuntu, command-line focused, though a GUI can be installed if needed.
Package Management	Uses PowerShell and the Windows Installer/MSI system, plus Windows Update for patches.	Uses apt for installing and updating software, which is simple and has a huge repository.	Uses dnf (or yum), the same package manager RHEL uses, which is known for being stable and enterprise focused.
Security	Built-in Windows Defender, Windows Firewall, and Active Directory integration for enterprise security policies.	Security updates through apt, AppArmor for access control, and a strong reputation for quick patching.	Security-Enhanced Linux (SELinux) by default, plus frequent updates since it tracks RHEL closely.
Performance	Heavier on system resources because of the GUI and background services, even in Server Core mode.	Lightweight runs well on minimal hardware since there's no GUI overhead.	Similarly lightweight, performance is close to Ubuntu since both are Linux based.
Typical Enterprise Use Cases	Active Directory domains, .NET applications, Microsoft Exchange, SQL Server, environments already built around Microsoft tools.	Web hosting, cloud infrastructure, Docker/Kubernetes, DevOps pipelines, startups and cloud-native companies.	Enterprise environments that need RHEL-like stability without paying for RHEL, common in banking, government, and long-term production systems.

Advantages	Easy to use for Windows-familiar admins, strong enterprise support, tight integration with Microsoft products.	Free, huge community support, frequent updates, works great for cloud and container workloads.	Very stable, long support lifecycle, good for organizations that need predictability over the newest features.
Disadvantages	Expensive licensing, heavier resource usage, more attack surface due to the GUI.	Not ideal for organizations already built around Microsoft software.	Smaller community than Ubuntu, slower to get bleeding-edge software versions.

Problems Encountered

- During sudo apt upgrade, the package linux-firmware-amd-misc repeatedly failed to download from the Philippines Ubuntu mirror (403 Forbidden). This did not affect server functionality, since the package is optional firmware not required for the VM.
- The "Install OpenSSH server" checkbox during installation did not apply correctly. Running systemctl status ssh after logging in returned "Unit ssh.service could not be found."

Solutions Implemented

- Re-ran sudo apt update and sudo apt full-upgrade -y --fix-missing to complete the update while skipping the unavailable firmware package.
- Manually installed and enabled SSH post-installation with sudo apt install openssh-server -y, followed by sudo systemctl enable ssh and sudo systemctl start ssh. Verified with systemctl status ssh, which then showed active (running).

Lessons Learned

Installing Ubuntu Server demonstrated the importance of careful system configuration, documentation, and verification before deploying an enterprise server.

References

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